

ZERO CHASSIS OUTPUT REFERENCE MANUAL

A practical guide to every calculated parameter — what it means, why it matters, and what moves when you change it

Independent documentation · Written against Zero Chassis v3.8.x · v1.0, July 2026

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How Zero Chassis outputs hang together

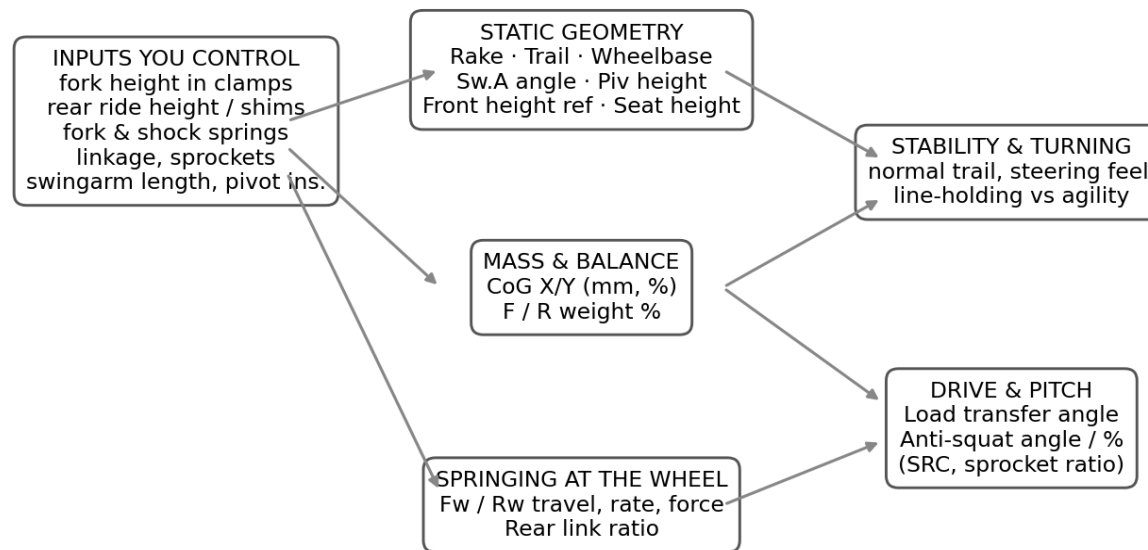


Fig. 0 — The big picture: what you adjust on the bike flows through geometry, mass and springing into the two things that decide lap time: stability/turning and drive/pitch control.

This is independent, unofficial documentation and is not affiliated with or endorsed by Zero Chassis Software. Parameter definitions were verified against the software's own tooltips and numerically against a real setup file (§8).

Acknowledgements

With thanks to community reviewers, in particular Jake Skate, for the wheelbase/drive-traction correction, extended-position swingarm-angle reality check, and the front height reference datum.

1. How to use this manual

Every value Zero Chassis prints falls into one of five groups: static geometry, reference heights, suspension travel, rates & forces at the wheel, and acceleration (anti-squat) geometry. Each section defines the parameters in that group, gives a consistent worked example, explains why the value matters, and — in the final column — lists what else moves when you change it. That last column is the heart of setup work: on a motorcycle almost nothing changes in isolation.

Rules that govern everything in this document:

- Zero Chassis is a comparison tool, not an oracle. The absolute numbers matter less than the delta between the reference setup and a proposal. Always load the current, measured bike as [Ref] before modelling a change.
- Every screen value is computed for the position and lean shown in the header (e.g., Pos: TRACK, Lean: 0). The same bike at full lean or under brakes has different trail, wheelbase and anti-squat — static numbers are the starting point, not the whole story.
- Conventions: any value with a % sign is a percentage of wheelbase. Longitudinal positions (CoG X, SRC, SFC) are measured along X from the front wheel contact point. Component values (fork, shock) are what you measure in the workshop; wheel values (Fw, Rw) are what the tyre experiences.

Cross-checks. Section 8 shows the arithmetic that ties the values together, verified against the worked example. If you doubt a reading, run those checks — they confirm both the software's conventions and your input file.

2. Steering geometry

Steering geometry — rake, trail, normal trail

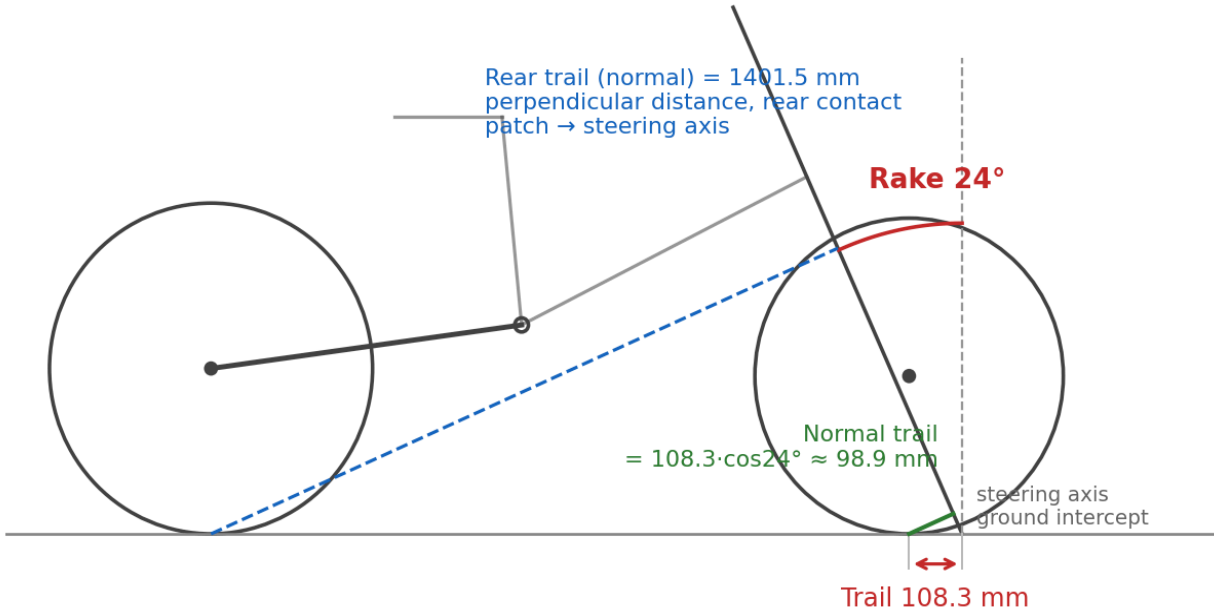


Fig. 1 — Rake, ground trail, normal trail, and rear normal trail. The steering axis intersects the ground ahead of the front contact patch.

Parameter	Example*	What it is	Why it matters	Knock-on effects when you change it
Rake	24°	Angle of the steering axis from vertical.	More rake = more stability, slower steering. Adjusted via front/rear ride heights, fork position in the clamps, or triple-clamp changes.	Trail moves with it (several mm per degree); front reference height, weight split and wheelbase shift slightly; anti-squat changes through chassis attitude. A 'rake change' is never only a rake change.
Trail	108.3 mm	Ground distance from the steering-axis ground intercept back to the front contact patch.	The primary stability number. More trail = stronger self-centring and stability on brakes; less = quicker turn-in but more nervous. Superbike window is roughly 95–120 mm.	Changing triple-clamp offset moves trail without moving rake — a surgical stability tool with no mass or drive-side effects. Changing rake or ride height moves trail and everything attitude-related together.
Rear Trail (normal)	1401.5 mm	Perpendicular distance from the rear contact patch to the steering axis.	Governs how strongly rear tyre lateral force steers the bike (rear-steer effect).	Derived — it follows rake, trail and wheelbase. Not adjusted directly; watch its delta when changing any of those, since large shifts alter how the rear influences steering.

Parameter	Example*	What it is	Why it matters	Knock-on effects when you change it
Wheelbase	1426 mm	Horizontal distance between contact patches at the current attitude.	Longer = stability and a higher wheelie threshold (more usable throttle when wheelie-limited) — but a lower rear load share, so in grip-limited drive, traction falls. Which regime you are in decides which way this lever helps. Adjusted with chain length / rear axle position.	Moving the rear axle also changes swingarm angle (and therefore anti-squat), rear weight %, load-transfer angle, and chain geometry. Sprocket changes move wheelbase as a side effect.
Sw.A angle	−7.8° laden / −12.4° extended	Swingarm angle from horizontal, pivot above axle (shown negative in ZC's sign convention). Strongly position-dependent: expect roughly 11–13° at full extension on a superbike, flattening to ~7–9° at laden track position. A companion row, Sw.A angle (frame ref), reports the same angle measured against the frame reference instead of the ground.	The single biggest lever on anti-squat. Always quote it with the position it was read at.	Moves with every rear ride-height, axle-position or pivot change — and drags anti-squat with it. It also tips chassis attitude, so rake and trail shift in sympathy.
Lean angle	0	The lean at which all values are computed.	Trail, effective wheelbase and anti-squat change significantly with lean.	A display setting, not a bike change. Re-run the sheet at expected mid-corner lean before signing off a change that targets corner behaviour — the static picture can look fine while the leaned picture is not.

* Example values throughout are one consistent worked dataset from a representative 1000 cc sportsbike, so the cross-checks in §8 can be reproduced.

3. Reference heights & chassis layout

Chassis layout — wheelbase, swingarm, pivot

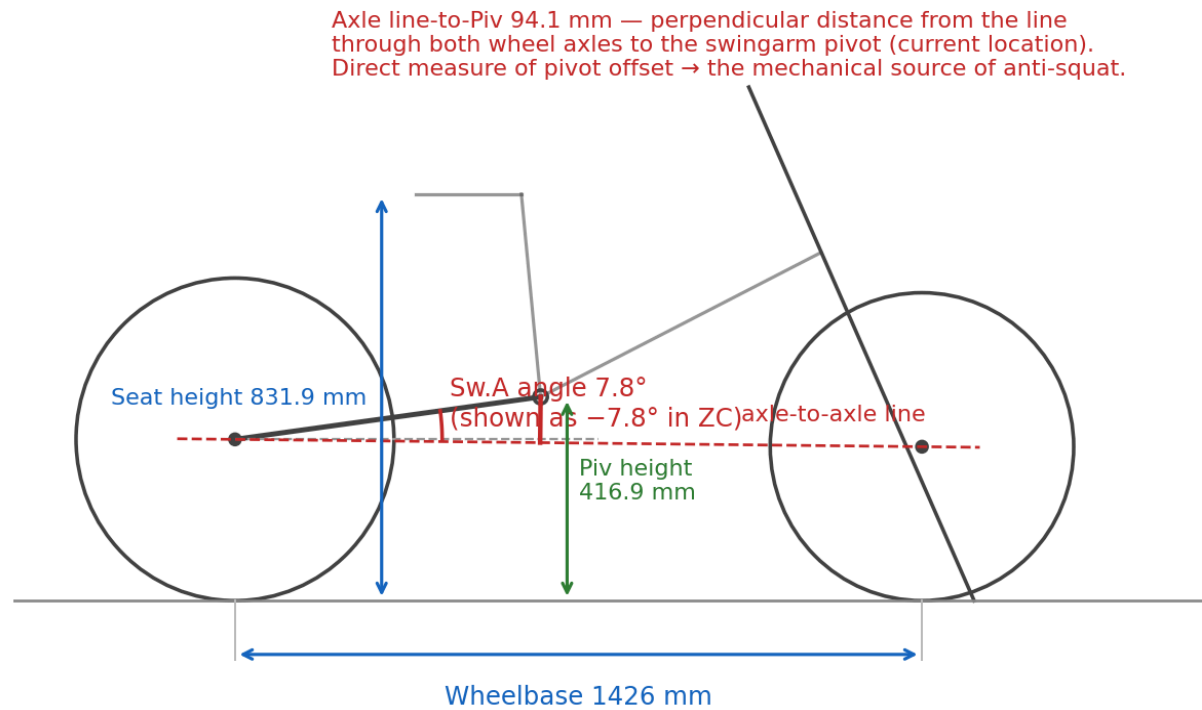


Fig. 2 — Wheelbase, swingarm angle, pivot height, seat height and the axle-line-to-pivot offset.

Parameter	Example*	What it is	Why it matters	Knock-on effects when you change it
Front height ref	733 mm	Height from the ground to the centre of the lower head-pipe (headstock) face — an official, fixed convention. That point is chosen because it stays constant even when triple clamps are swapped, so the reference survives front-end changes that would invalidate any clamp-based measurement.	A repeatable handle on front ride height that captures tyre and stroke position, unlike raw fork-height numbers.	Lowering the front steepens rake, reduces trail, adds front weight %, lowers the CoG slightly and nudges anti-squat through attitude — the standard 'more front grip, less stability' trade.
Seat height	831.9 mm	Height of the seat reference point above ground. The measurement point on the seat varies between bike files (e.g., one file records 'seat point taken at front of seat'),	Proxy for rear ride height and rider CoG placement.	Raising the rear steepens rake and cuts trail (quicker steering), increases swingarm angle and anti-squat, raises the CoG and shifts weight slightly forward — five effects from one shim.

Parameter	Example*	What it is	Why it matters	Knock-on effects when you change it
		so track the CHANGE in this value rather than comparing absolutes across bikes.		
Rear RH Tool	0	Reading from a rear ride-height measuring tool defined in the frame file (enabled with a Tool Y and tool angle in pivot-origin coordinates; a Ride Height Tool calibrate feature under Tools can derive these from two or more known readings — three recommended). 0 = tool not in use.	Maps a physical workshop ride-height tool measurement directly into the model, so shop numbers and model numbers reconcile without conversion.	Measurement bookkeeping only — no effect on the bike model unless used to set ride height.
Piv height	416.9 mm	Swingarm pivot height above ground.	With swingarm length, sets swingarm angle — so it drives anti-squat.	Changes swingarm angle and anti-squat, and alters chain run and linkage operating angles. Moves with ride height as well as with pivot inserts.
Axle line-to-Piv	94.1 mm	Perpendicular distance from the line through both wheel axles to the swingarm pivot at its current location (per the ZC tooltip).	The mechanical source of swingarm angle; referenced to the axle line rather than the ground, so it compares honestly across different ride heights.	Pivot inserts change it in isolation — anti-squat moves while ride height, CoG and attitude stay put. That separation is exactly why pivot inserts exist: ride-height changes can never move anti-squat alone. Note the on-screen label's (current) suffix: the value is position-dependent and grows as the suspension extends (e.g., 137.8 mm at full extension in the second worked dataset) — compare setups at the same position.

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4. Suspension travel

Travel and motion ratios — component values vs at-the-wheel values

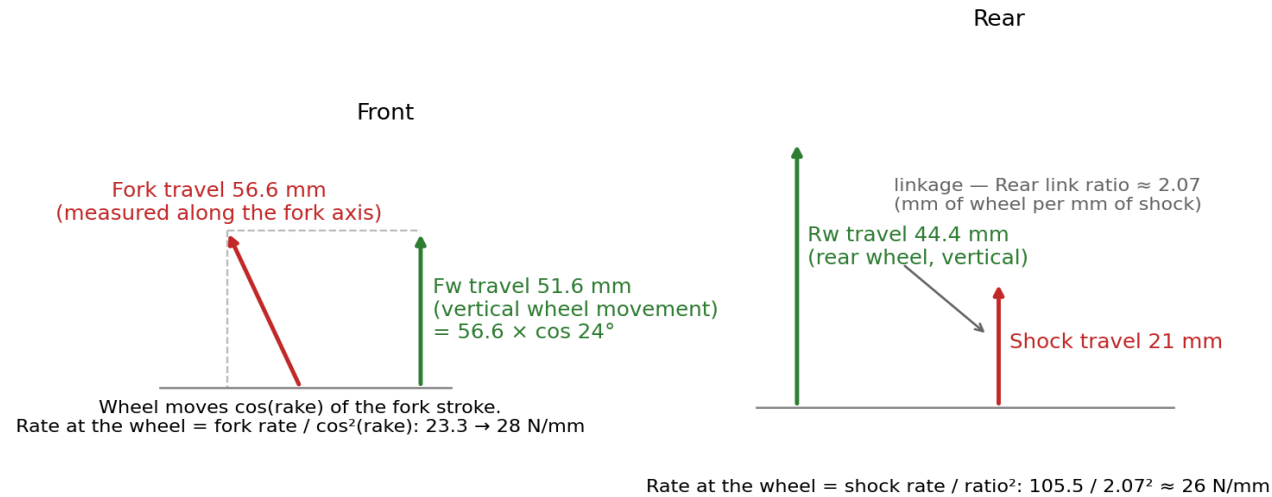


Fig. 3 — Component travel vs travel at the wheel. Front converts through $\cos(\text{rake})$; rear converts through the linkage ratio.

Zero Chassis reports travel twice: at the component (fork leg, shock shaft) and at the wheel (vertical). Component numbers match workshop measurements and travel sensors; wheel numbers are what the tyre experiences.

Parameter	Example*	What it is	Why it matters	Knock-on effects when you change it
Fork travel	56.6 mm	Stroke used along the fork axis at the current position, from full extension.	Directly comparable with fork potentiometer data — same axis, same convention.	A position output, not an adjustment. Preload, spring rate, oil level and load change where in the stroke the bike sits — and every position-dependent value (rake, trail, anti-squat) moves with it.
Shock travel	21 mm	Shock shaft stroke used at the current position.	Matches the shock potentiometer. Small numbers are normal — the linkage multiplies them at the wheel.	As above, rear side: preload and spring changes move the operating position, which moves swingarm angle and anti-squat.
Fw travel	51.6 mm	Vertical front-wheel movement = fork travel $\times \cos(\text{rake})$.	The tyre's-eye view of front stroke; use it when comparing front/rear balance in common units.	Derived. A rake change alters the fork-to-wheel conversion, so identical fork stroke produces different wheel travel.
Rw travel	44.4 mm	Vertical rear-wheel movement (shock travel $\times$ linkage ratio).	Rear stroke in tyre terms.	Derived. A linkage change re-scales it even with the same shock stroke.

Parameter	Example*	What it is	Why it matters	Knock-on effects when you change it
F/R tyre travel	0 / 0	Tyre carcass deflection included in the model (0 = rigid-tyre model).	With tyre rates entered, part of the 'suspension' happens in the carcass.	Entering tyre spring rates redistributes modelled travel between carcass and suspension — total wheel movement stays honest but the split changes, which matters when correlating to sensor data.
F/R tyre rad (effective)	304.6 / 335.9 mm	Effective rolling radius of each tyre used by the model, from the tyre files.	Sets contact-patch positions, so it underpins wheelbase, trail, CoG height and the anti-squat construction.	Changes with tyre choice, pressure and wear assumptions in the tyre file. A tyre swap that changes effective radius is a ride-height and geometry change the model will show — and the workshop will not.

* Example values throughout are one consistent worked dataset from a representative 1000 cc sportsbike, so the cross-checks in §8 can be reproduced.

5. Rates & forces at the wheel

Same logic as travel: the spring you buy is not the spring the tyre feels. Geometry transforms both stiffness and force.

Parameter	Example*	What it is	Why it matters	Knock-on effects when you change it
Fork rate	23.3 N/mm	Combined front spring rate along the fork axis (both legs).	The workshop-facing number — what you physically change.	Moves Fw rate and force one-for-one, changes where the bike sits under load (hence attitude, rake, trail), stiffens pitch under brakes, and pulls SRC forward.
Shock rate	105.5 N/mm	Shock spring rate at the shaft.	The purchased rear spring.	Moves Rw rate through the linkage, changes rear operating position (hence swingarm angle and anti-squat), and pushes SRC rearward.
Fw rate	28 N/mm	Front wheel rate = fork rate ÷ cos ² (rake).	Vertical stiffness at the front tyre. Keep it constant through geometry changes unless a spring change is intended.	Derived — but note it changes when rake changes with no spring swap at all. A 1° rake change is a small hidden spring change; ZC makes it visible.
Rw rate	26 N/mm	Rear wheel rate = shock rate ÷ (link ratio) ² .	Vertical stiffness at the rear tyre. Front/rear wheel-rate balance is the real 'spring balance' of the bike.	Derived — moved by either the spring or the linkage. Because the ratio enters squared, small linkage differences produce large wheel-rate differences.
Fork force	1542.4 N	Spring force in the fork at the current position (preload + compression).	With Fw force, shows where in the stroke the bike sits — the modelled equivalent of sag.	A position output. Preload changes force at a given position; spring changes alter how force builds with stroke.
Shock force	3717.3 N	Spring force in the shock at the current position.	Divided by the link ratio it becomes vertical force at the rear tyre.	As above, rear side.
Fw force	1692.3 N	Vertical spring force at the front wheel = fork force ÷ cos(rake).	≈ the mass supported at the front contact patch — sanity-check against corner-weight scales.	Follows load placement and front spring/preload. If it disagrees with the scales, the bike file or the measured sag is wrong.
Rw force	1798.4 N	Vertical spring force at the rear wheel = shock force ÷ link ratio.	≈ mass at the rear patch; front + rear should reconcile with bike + rider weight.	As above, rear side.

Parameter	Example*	What it is	Why it matters	Knock-on effects when you change it
Rear link ratio	2.07	Instantaneous linkage motion ratio: the slope of the wheel-travel vs shock-travel curve over a small increment around the current point (officially defined this way — which is why it differs from simply dividing current wheel travel by current shock travel).	Rises or falls through the stroke depending on linkage design (progression).	A linkage change re-scales R_w travel, rate and force simultaneously, changes progression, and alters effective damping at the wheel — it usually demands a compensating spring-rate change. Model before buying parts.
SRC mm / %	686.8 / 48.16%	Spring rate centre: the longitudinal position where combined wheel-rate stiffness is centred, measured along X from the front contact point; % of wheelbase. $SRC = R_w \text{ rate} \div (F_w \text{ rate} + R_w \text{ rate}) \times WB$.	Compare directly with CoG X. SRC ahead of CoG X = front stiffer relative to its load share (rear settles more under combined load); behind = the reverse. Matching them balances pitch stiffness against mass.	Every spring or linkage change moves it. Check its shift against CoG X after any spring change — it tells you whether you also changed the bike's pitch balance, not just its stiffness.
SFC mm / %	—	Spring force centre: the balance point of the current front and rear wheel FORCES, measured along X from the front contact point; % of wheelbase. Force-based sibling of SRC.	SRC describes stiffness balance; SFC describes where load is actually carried right now. In a static position SFC should sit near CoG X — divergence indicates preload imbalance between the ends.	Moves with preload, springs, and position; unlike SRC it changes through the stroke even with fixed springs, because forces build with travel.

* Example values throughout are one consistent worked dataset from a representative 1000 cc sportsbike, so the cross-checks in §8 can be reproduced.

6. Mass & centre of gravity

Centre of gravity and static weight distribution

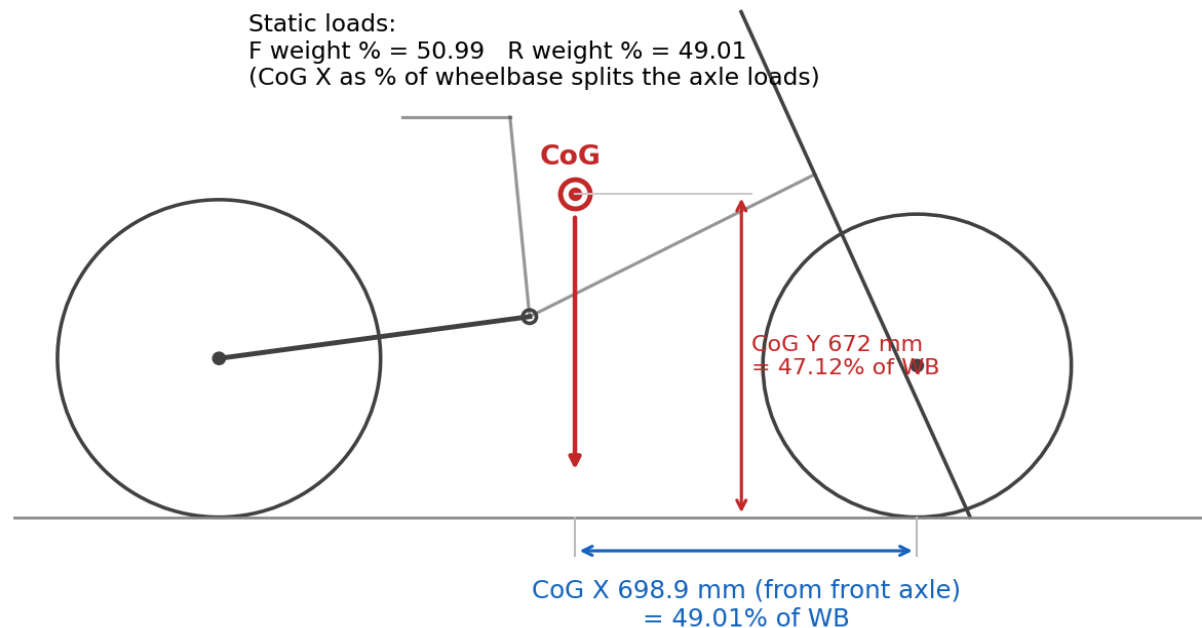


Fig. 4 — CoG location and static weight split. CoG X is measured from the front axle; heights are expressed in mm and as % of wheelbase.

Parameter	Example*	What it is	Why it matters	Knock-on effects when you change it
CoG X mm / %	698.9 / 49.01%	Horizontal position of the combined bike + rider (fully laden) CoG, measured from the front wheel contact point; % of wheelbase. (At zero lean the contact point sits below the axle, so the X value matches an axle-referenced measurement — at lean they differ.)	Sets static weight distribution. Provenance note: the underlying CoG lives in the frame file in pivot-origin coordinates and is only as good as its source — many files carry an estimated CoG. Use the COG Wizard (Tools menu) or measured data for setup-critical work; the Separate entry method (bike/rider/fuel masses) is required for the Model calculation type.	Moving rider position, ballast, or major components moves it. Weight %s follow directly; wheelie and stoppie thresholds shift; anti-squat demand changes slightly.

Parameter	Example*	What it is	Why it matters	Knock-on effects when you change it
CoG Y mm / %	672 / 47.12%	CoG height above ground; % is height ÷ wheelbase.	The master number for load transfer under braking and acceleration.	Raising it increases the load-transfer angle — which lowers anti-squat % even with zero drive-side change — and deepens pitch under brakes and throttle. Ride-height changes move it as a side effect.
F weight %	50.99%	Static front axle load share.	Front-biased setups aid turn-in and front confidence; too much starves rear drive grip.	Derived from CoG X. Anything that moves mass or ride height moves it; check it after 'geometry-only' changes to catch unintended balance shifts.
R weight %	49.01%	Static rear axle load share (equals CoG X % from the front axle).	The launch and corner-exit tyre's share. F% + R% = 100 — a built-in consistency check.	As above.

* Example values throughout are one consistent worked dataset from a representative 1000 cc sportsbike, so the cross-checks in §8 can be reproduced.

7. Acceleration geometry — anti-squat

Acceleration geometry — anti-squat construction

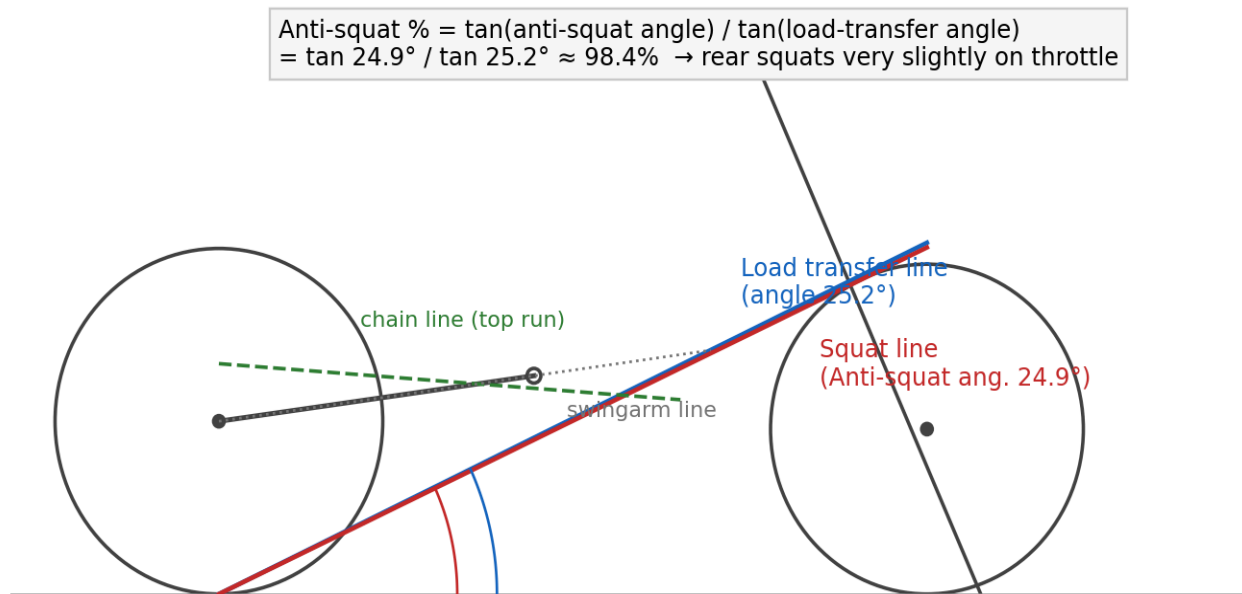


Fig. 5 — The classic construction: the squat line (rear contact patch through the chain-line/swingarm-line intersection) versus the load-transfer line (slope set by CoG height over wheelbase).

Under throttle, load transfers rearward and wants to compress the shock ('squat'). Simultaneously, chain pull and the driving force acting through the swingarm try to extend it. Anti-squat is the ratio of those effects.

Parameter	Example*	What it is	Why it matters	Knock-on effects when you change it
Load tran ang.	25.2°	Angle of the load-transfer line: $\tan(\text{angle}) = \text{CoG height} \div \text{wheelbase}$.	The 'demand' side: how hard acceleration tries to squat the rear. Purely a mass-property number.	Only mass placement moves it — CoG height, CoG position, wheelbase. If it changes, anti-squat % changes even with the drivetrain geometry untouched: adding a taller rider is an anti-squat change.
Anti-squat ang.	24.9°	Angle of the anti-squat line: from the rear contact point to the Instantaneous Force Centre (IFC) — the intersection of the	The 'supply' side: how hard the drivetrain resists squat.	Rear ride height, pivot position, swingarm length and sprocket sizes all move it. Falls as the suspension compresses (the swingarm flattens), so quote it with the position it was read at.

Parameter	Example*	What it is	Why it matters	Knock-on effects when you change it
		swingarm line (pivot to rear axle) and the top chain run.		
Anti-squat %	98.4%	$\tan(\text{anti-squat angle}) \div \tan(\text{load-transfer angle}) \times 100$.	100% = rear holds its height under throttle; <100% squats; >100% extends. Squat lowers the CoG and calms wheelies but runs wide and loads the tyre edge; extension holds geometry but can feel harsh and break traction on bumps. Superbikes typically tune just under to just over 100% per circuit.	Depends on the CoG location — with an unverified (estimated) CoG, prefer tracking the anti-squat ANGLE instead, which uses pure geometry and no CoG. Not constant through travel: the second worked dataset reads 117.4% at full extension, and the same physics takes a laden bike toward or below 100% as the swingarm flattens through the stroke — use the graph view across travel, not the single static number.
Accel squat/extend	Extend	Categorical flag stating the direction of rear suspension movement under acceleration implied by the anti-squat %.	Quick-read companion to the %: Extend when anti-squat > 100%, Squat when < 100%.	Follows anti-squat % directly; no independent adjustment.
Sprocket ratio	2.8	Rear ÷ front sprocket teeth (e.g., 42/15).	Sets gearing — and the chain-line angle that feeds anti-squat.	A 'gearing change' is also an anti-squat change (chain line) and a wheelbase change (chain length). One tooth on the front moves anti-squat noticeably. Always re-run the model after sprocket changes.

* Example values throughout are one consistent worked dataset from a representative 1000 cc sportsbike, so the cross-checks in §8 can be reproduced.

8. Cross-checks — proving the numbers tie together

These identities hold in the worked-example dataset used throughout this manual. Run them whenever you build a new bike file or doubt an input; if one fails, an input is wrong.

- Front travel: Fw travel = fork travel $\times \cos(\text{rake}) \rightarrow 56.6 \times \cos 24^\circ = 51.7 \approx 51.6$ ✓
- Front rate: Fw rate = fork rate $\div \cos^2(\text{rake}) \rightarrow 23.3 / 0.834 = 27.9 \approx 28$ ✓
- Front force: Fw force = fork force $\div \cos(\text{rake}) \rightarrow 1542.4 / 0.9135 = 1688 \approx 1692$ ✓
- Rear force: Rw force = shock force $\div \text{link ratio} \rightarrow 3717.3 / 2.07 = 1796 \approx 1798$ ✓
- Rear rate: Rw rate = shock rate $\div \text{ratio}^2 \rightarrow 105.5 / 26 = 4.06$, $\sqrt{4.06} = 2.01$ (instantaneous ratio; the displayed 2.07 is the average — a small difference is expected)
- Rear normal trail: $(\text{WB} + \text{trail}) \times \cos(\text{rake}) \rightarrow 1534.3 \times \cos 24^\circ = 1401.7 \approx 1401.5$ ✓
- Load transfer angle: $\text{atan}(\text{CoG Y} \div \text{WB}) \rightarrow \text{atan}(672 / 1426) = 25.2^\circ$ ✓
- Anti-squat %: $\tan 24.9^\circ \div \tan 25.2^\circ = 98.6\% \approx 98.4\%$ ✓ (rounding of displayed angles)
- Weight split: R weight % = $\text{CoG X} \div \text{WB} = 698.9 / 1426 = 49.01\%$ ✓ ; F% = $100 - 49.01 = 50.99\%$ ✓
- Spring rate centre: $\text{SRC} = \text{Rw rate} \div (\text{Fw rate} + \text{Rw rate}) \times \text{WB} \rightarrow 26 / 54 \times 1426 = 686.6 \approx 686.8$ ✓ (48.15% $\approx$ 48.16%)

The same identities at a different position (second worked dataset, Pos: Ext)

A second dataset — a different setup of the same model, read at full extension (Pos: Ext, Lean 0) — confirms the formulas hold at any position and reveals behaviour the laden view hides: WB 1424.9, rake 24.3°, trail 110.3, Sw.A -12.4°, CoG 723/717, Fw rate 38.7, Rw rate 55.9, shock rate 250.1, link ratio 2.12, anti-squat 117.4%.

- Rear normal trail: $(1424.9 + 110.3) \times \cos 24.3^\circ = 1399.2 \approx 1398.9$ ✓
- Load transfer angle: $\text{atan}(717 / 1424.9) = 26.7^\circ$ ✓ ; anti-squat %: $\tan 30.6^\circ / \tan 26.7^\circ = 117.6\% \approx 117.4\%$ ✓
- Weight split: $723 / 1424.9 = 50.74\% = \text{R weight \%}$ ✓ ; SRC: $55.9 / 94.6 \times 1424.9 = 842.0 \approx 842.1$ (59.1%) ✓
- Top-out springs are in the numbers: shock rate at full extension = main spring + rear top-out spring $\rightarrow 100 + 150$ (TOS 150x8) = $250 \approx 250.1$ ✓ . Rw rate follows: $250.1 / 2.12^2 = 55.6 \approx 55.9$ ✓
- Negative fork force (-34.2 N) and Fw force (-37.5 N) at full extension are the top-out spring region — the spring is holding the wheel, not supporting the bike. Rates and forces displayed near full droop include the engaged top-out springs; do not read them as main-spring values.

9. Applying it — setup development workflow

The process

- 1. Baseline: measure the bike, build/verify the ZC bike file, save the current setup as [Ref]. Run the §8 cross-checks.
- 2. Define the problem in tyre/chassis terms from rider comment plus data (e.g., 'rear breaks traction over exit kerbs', not 'needs more grip').
- 3. Model the candidate change in a comparison setup and read the deltas — especially the ones you did NOT intend. The knock-on columns above exist because every ride-height change moves rake, trail, wheelbase, CoG and anti-squat together.
- 4. Change one thing at the track, keep wheel rates constant unless springs are the change, and log the ZC setup file name against each session.
- 5. Correlate: suspension potentiometers against modelled travel, corner weights against Fw/Rw force, and rider comment against the geometry delta.

Symptom → parameter map

Symptom / goal	Parameters to look at (and typical direction)
Slow steering / won't finish the corner	Trail (reduce), rake (steepen via rear ride height up or fork position in clamps), wheelbase (shorten). Watch: anti-squat rises when the rear goes up.
Nervous on brakes / headshake on exit	Trail (increase), wheelbase (lengthen), steering damper last. Check front ride height isn't collapsing trail under brakes.
Rear squats and runs wide on throttle	Anti-squat % (raise): rear ride height up, pivot lower relative to the axle line, or smaller rear / larger front sprocket. Confirm the % across travel, not just static.
Rear harsh / chatters on exit bumps	Anti-squat % may be too high — reduce it; or Rw rate too high — check the wheel rate, not just the spring. Consider the linkage curve.
Wheelies too early, killing drive	CoG Y (lower), CoG X (forward), anti-squat (slightly reduce so squat drops the CoG), sprocket ratio.
Poor drive grip off slow corners	R weight % (increase via rider/ballast), anti-squat toward ~100%, Rw rate (soften at the wheel), check tyre-travel assumptions.
Front vague mid-corner	F weight %, front ride height (via the front height reference), Fw rate. Re-check at Lean $\neq 0$ — the static picture can look fine.
After ANY sprocket change	Re-run the model: wheelbase and anti-squat both moved even though you 'only changed gearing'.

10. Conventions & verification notes

- Any value with a % sign is a percentage of wheelbase (confirmed in-software).
- SRC = spring rate centre; Axle line-to-Piv = perpendicular distance from the line through both wheel axles to the swingarm pivot at its current location (both confirmed from the software's tooltips).
- The front height reference datum is defined per bike file — record which physical point (e.g., bottom yoke, headstock) your files use, or setups stop being comparable between bikes.

- Everything else is standard motorcycle-dynamics convention, cross-checked against the official Zero Chassis help file, and verified numerically against real setup files in §8.

Bike position buttons

All results are position-dependent, and the position buttons control that position. Ext (the default) is fully extended: 0 mm fork and shock travel. Every other preset applies fixed VERTICAL FORCES at the wheel contact points rather than fixed travel — Static/1g: 950/950 N, Acceleration: 480/1450 N, Braking: 2900/0 N, Cornering: 1850/1600 N, Cornering-with-lean adds 50° of lean. Because the input is force, any setup change shows up as a change in resulting travel and attitude — that is the analytical point: with constant applied forces, every movement of the numbers is caused by your change, not by the loading. Custom positions (e.g., a track-specific suspension position) can also be saved as presets.

The frame file & coordinate system

- Origin: the XY origin of the frame file is the swingarm pivot axis (centre of the pivot hole). For adjustable pivots, the zero-adjustment position is (0,0). Positive X points toward the REAR wheel; positive Y points UP. The XY plane is the frame's centre (symmetry) plane.
- The frame file stores an HP_Angle (nominal head angle, e.g., 24°) as the alignment reference for editable coordinates such as CoG and the rear ride-height tool.
- Head-pipe insert geometry can be defined as 'upper and lower' points or as 'offset and angle' about the upper, centre, or lower pivot, with a normal/reversed angle-direction convention. All head-pipe measurements in a file (HP_Triple_Top/Bottom, HP_Length, insert numbers) must reference the same physical point convention.
- The engine file must use the same coordinate system as its frame file.
- Best practice: record datum choices (head-pipe point convention, seat point, CoG provenance) in the frame file's Description field, so every setup built on that file is interpretable later.